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1. File Operations

a. Creating and navigating directory

1. **ls**: This command helps us to see all files and folders.

```
Ujwal0608@Ubuntu:~/Desktop$ ls  
c_folder  my_os_folder  newfile.txt  test  test.c
```

Figure 1: Use of ls

2. **mkdir**: It is used to create a new directory/folder.

```
Ujwal0608@Ubuntu:~/Desktop$ mkdir linux_lab
```

Figure 2: Use of mkdir

3. **cd**: It is used to navigate into the given directory.

```
Ujwal0608@Ubuntu:~/Desktop$ cd linux_lab  
Ujwal0608@Ubuntu:~/Desktop/linux_lab$
```

Figure 3 : Use of cd

4. **nano**: It is used to create a file and open a text editor. We can just type the data inside that file and then save:
 - i. Ctrl + o and then click Enter (to save first)
 - ii. Ctrl + x (to exit)

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ nano greeting.txt
```

Figure 4 : Use of nano

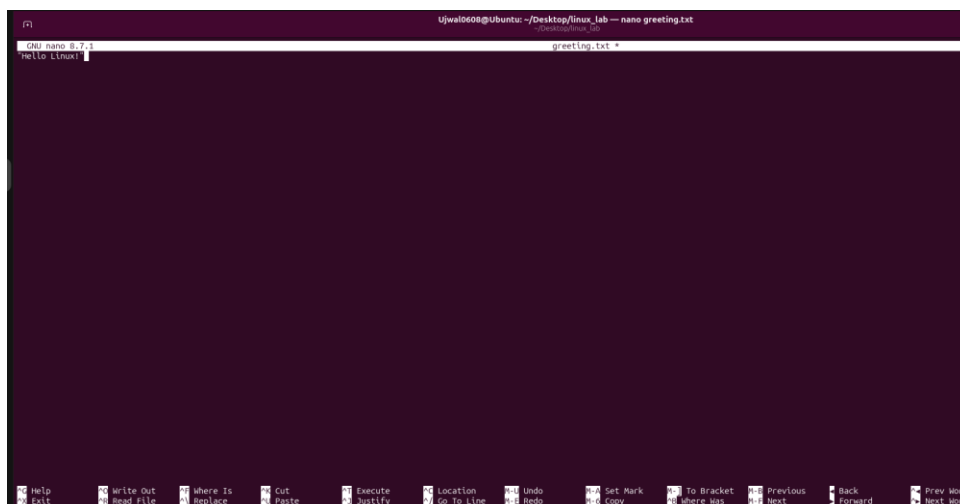


Figure 5: Text editor using nano

5. **cat**: to view file data inside the given file.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ cat greeting.txt
"Hello Linux!"
Ujwal0608@Ubuntu:~/Desktop/linux_lab$
```

Figure 6: Use of cat

6. **cp**: to copy the content of a file into a new file.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ cp greeting.txt backup.txt
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
backup.txt  greeting.txt
```

Figure 7: Use of cp

7. **mv**: This command is used to rename a file.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ mv backup.txt old_greeting.txt
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
greeting.txt  old_greeting.txt
```

Figure 8: Use of mv

8. **rm**: to remove files.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ rm old_greeting.txt
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
greeting.txt
```

Figure 9: Use of rm

9. **rmdir**: It is used to empty remove directory.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ mkdir temp_dir
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
greeting.txt  temp_dir
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ rmdir temp_dir
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
greeting.txt
```

Figure 10: Use of rmdir

10. **rm -r**: It is used to delete a non-empty directory.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
greeting.txt
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ mkdir temp_dir
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ cd temp_dir
Ujwal0608@Ubuntu:~/Desktop/linux_lab/temp_dir$ touch test.txt
Ujwal0608@Ubuntu:~/Desktop/linux_lab/temp_dir$ ls
test.txt
Ujwal0608@Ubuntu:~/Desktop/linux_lab/temp_dir$ cd ..
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
greeting.txt  temp_dir
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ rm -r temp_dir
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls
greeting.txt
Ujwal0608@Ubuntu:~/Desktop/linux_lab$
```

Figure 11: Use of rm -r

11. **ls -l**: It is used to check permissions.

```
Ujwal0608@ubuntu:~/Desktop/linux_lab$ ls -l greeting.txt  
-rw-r--r-- 1 Ujwal0608 imposter_gorkha 15 Sep  9 09:17 greeting.txt
```

Figure 12: Use of ls -l

2. APT package management

- a. **sudo apt update:** used to update package information.

```
ujwal0608@Ubuntu:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://np.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://np.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://np.archive.ubuntu.com/ubuntu resolute-backports InRelease
31 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Figure 13: Use of sudo apt update

- b. **sudo apt install build-essential:** used to install essential tools needed to compile and build programs on Ubuntu.

```
ujwal0608@Ubuntu:~$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 31
```

Figure 14: Use of sudo apt install build-essential

- c. **Verifying installation:** used to verify the installation.

```
ujwal0608@Ubuntu:~/Desktop$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Figure 15: Verify the compiler version

- d. **Search for other packages:** used to search other Python packages.

```
Ujwal0608@Ubuntu:~/Desktop$ apt search python3
alembic/resolute 1.18.4-1 all
lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
Computer-aided design CAD tool

astro-frameworks/resolute 5.0 all
Astronomical data analysis frameworks

astro-python3/resolute 5.0 all
Python 3 packages for astronomy

bamkit/resolute 0.0.1+git20170413.ccd079d-3build1 all
tools for common BAM file manipulations

beancount/resolute 3.2.0-2build1 all
Double-entry accounting from text files

berrynet-dashboard/resolute 3.10.2-2build1 all
deep learning gateway - python3 modules

brlTTY/resolute,now 6.7-1ubuntu6 amd64 [installed,automatic]
Access software for a blind person using a braille display

budgie-app-launcher-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to provide an alternative means to launch applications

budgie-brightness-controller-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to control the brightness of the screen

budgie-clockworks-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to display clock across multiple time zones

budgie-countdown-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet providing a countdown capability on the Budgie Desktop

budgie-dropby-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
Applet to popup when a USB device is connected
```

Figure 16: Searching for Python packages

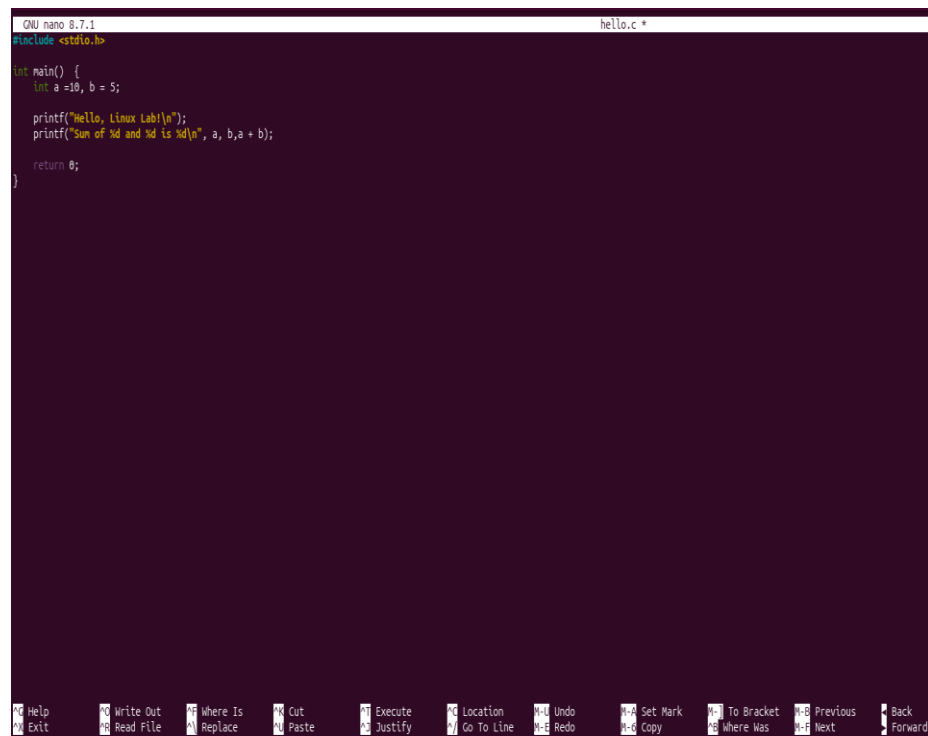
3. C Compilation

a. Step 1: Creating a C file and writing a program

- i. **nano <file_name>:** It is used to create a file and open a text editor to write file content.

```
Ujwal0608@Ubuntu:~/Desktop$ ls
c_folder  linux_lab  my_os_folder  newfile.txt  test  test.c
Ujwal0608@Ubuntu:~/Desktop$ cd linux_lab
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ nano hello.c
Ujwal0608@Ubuntu:~/Desktop/linux_lab$
```

Figure 17 : Use of nano



The screenshot shows the nano 2.7.1 text editor interface. The title bar indicates the file is 'hello.c *'. The editor contains the following C code:

```
#include <stdio.h>

int main() {
    int a = 10, b = 5;
    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);
    return 0;
}
```

The bottom status bar displays various keyboard shortcuts for editor functions: Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next, Back, and Forward.

Figure 18: Inside the text editor using nano

b. Step 2: Compilation and Run

- i. **gcc <filename> -o <executable_name>**: it specifies the output in executable format.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ gcc hello.c -o hello_program
```

Figure 19: Convert into executable form

- ii. **Ls -l <executable_name>**: it checks whether the executable was created or not.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ls -l hello_program
-rwxr-xr-x 1 Ujwal0608 imposter_gorkha 15992 Sep  9 09:54 hello_program
```

Figure 20: Checking executable form

- iii. **./<executable_name>**: It runs the compiled program.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

Figure 21: Executing the program

- iv. **gcc -Wall <file_name> -o <compile_name with warnings>**: It is used to compile the program with warnings enabled.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ gcc -Wall hello.c -o hello_program_warn
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

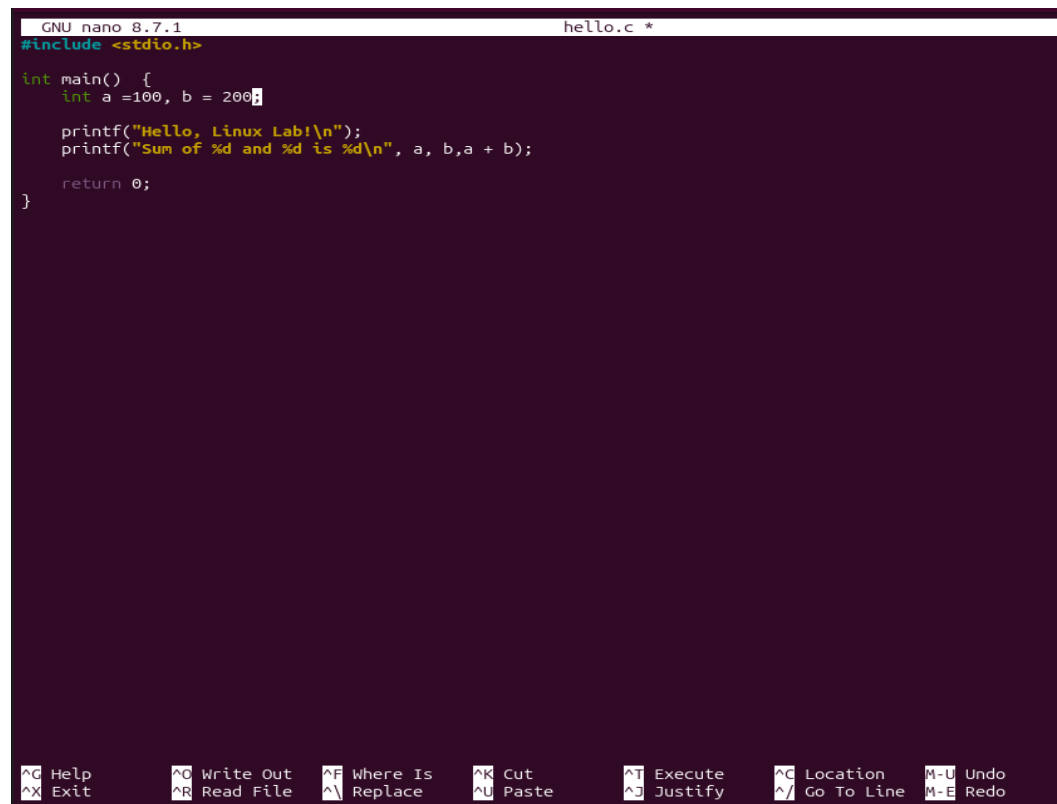
Figure 22: Compiling program with warnings

c. Step 3: Modifying and Recompiling

- i. **nano <file_name>:** used to create a file and open a text editor to edit previous file content.

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ nano hello.c
```

Figure 23: Use of nano



```
GNU nano 8.7.1 hello.c *
#include <stdio.h>

int main() {
    int a =100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b,a + b);

    return 0;
}
```

Figure 24: Inside the text editor using nano

- ii. **gcc -Wall <file_name> -o <executable_name> and ./<executable_name>:** used to execute the program

```
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ gcc -Wall hello.c -o hello_program
Ujwal0608@Ubuntu:~/Desktop/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 is 300
```

Figure 25: Compiling a program into executable form

4. Clean up and Verification

- a. **cd**: It is used to go back to the home directory

```
Ujwal0608@Ubuntu:~/Desktop$ cd
Ujwal0608@Ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
```

Figure 26: Use of cd

- b. **ls -lh <directory_name>**: it is used to list all files inside the given directory with details.

```
Ujwal0608@Ubuntu:~$ cd Desktop
Ujwal0608@Ubuntu:~/Desktop$ ls -lh linux_lab
total 40K
-rw-r--r-- 1 Ujwal0608 imposter_gorkha 15 Sep  9 09:17 greeting.txt
-rw-r--r-- 1 Ujwal0608 imposter_gorkha 164 Sep  9 10:00 hello.c
-rwxr-xr-x 1 Ujwal0608 imposter_gorkha 16K Sep  9 10:01 hello_program
-rwxr-xr-x 1 Ujwal0608 imposter_gorkha 16K Sep  9 09:55 hello_program_warn
```

Figure 27: Listing all files in details

- c. **cat <directory_name>/<file_name>**:view the content inside the given file .

```
Ujwal0608@Ubuntu:~/Desktop$ cat linux_lab/hello.c
#include <stdio.h>

int main() {
    int a =100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b,a + b);

    return 0;
}
```

Figure 28: Use of cat

- d. **rm -rf <directory_name>**:used to delete entire given directory .

```
Ujwal0608@Ubuntu:~/Desktop$ rm -rf linux_lab
Ujwal0608@Ubuntu:~/Desktop$ ls
c_folder  my_os_folder  newfile.txt  test  test.c
```

Figure 29: Deleting entire directory content

5. Summary

In general, using **ls** we can view all files and folders; using **mkdir** and **cd** we can create a new folder and able to navigate inside it; and to create a new file we use **nano**, which opens a text editor in which we can keep data. To view the file content, we use **cat**, and to copy that file content into a new file, we use **cp**; and to rename the file, we use **mv**.

To remove a file, we use **rm <file_name>**, and to remove the directory, we can use **rm -r dir**.

We also have to install necessary packages using **sudo apt update** and **sudo apt install package**. For compiling a C program, we have to first convert a C file into executable form using **gcc file.c -o output**, and then, after using **./output**, we can run our program.